SALES REVENUE, PROFIT MARGIN & MONTHLY GROWTH ANALYSIS

Week 6 – Final Data Analytics Project

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Course: B.Tech – Artificial Intelligence and Machine Learning

Project Tools: Python, Pandas, NumPy, Matplotlib, Power BI, Git and GitHub

1. INTRODUCTION

Data analytics plays an important role in understanding business performance and supporting decision-making. Organizations use sales data to identify profitable products, evaluate regional performance, monitor revenue trends, and improve business strategies.

The project titled **Sales Revenue, Profit Margin & Monthly Growth Analysis** focuses on analyzing sales transaction data using Python and Microsoft Power BI.

The project includes data preparation, exploratory data analysis (EDA), statistical analysis, data visualization, and the development of an interactive business intelligence dashboard.

2. PROJECT OBJECTIVES

The main objectives are:

1. Analyze total sales revenue and estimated profit.
2. Evaluate product-wise sales performance.
3. Compare revenue generated across different regions.
4. Analyze category-wise revenue distribution.
5. Calculate estimated profit margins.
6. Examine monthly revenue trends and growth.
7. Develop an interactive Power BI dashboard.
8. Maintain project files and version history using Git and GitHub.

3. TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Python	Data analysis and processing
Pandas	Data manipulation
NumPy	Numerical calculations
Matplotlib	Data visualization
Power BI	Interactive dashboard
Git	Version control
GitHub	Project hosting and documentation
CSV	Dataset storage

4. DATASET DESCRIPTION

The project uses an extended sales dataset containing 20 illustrative sales records.

The dataset includes the following fields:

- Order_ID
- Product
- Category
- Region
- Units_Sold
- Unit_Price
- Customer_Rating
- Revenue
- Order_Date
- Estimated_Cost_Rate
- Estimated_Cost
- Estimated_Profit
- Profit_Margin_Pct
- Year_Month
- Month
- Year

The original sales dataset was extended with illustrative dates and assumed product cost rates to support monthly and profitability analysis.

5. DATA PREPARATION

Data preparation was performed using Python and Pandas.

The following activities were completed:

1. Imported the CSV dataset.
2. Examined the dataset structure and column names.
3. Checked missing values.
4. Identified duplicate records.
5. Generated statistical summaries.
6. Added estimated cost and profit calculations.
7. Created date-related fields for monthly analysis.
8. Prepared the dataset for visualization and Power BI.

6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify useful business patterns.

The analysis included:

- Dataset shape and structure.
- Missing-value analysis.
- Duplicate-record analysis.
- Statistical summary.
- Product-wise revenue analysis.
- Region-wise revenue analysis.
- Category-wise revenue analysis.
- Monthly revenue and growth analysis.
- Estimated profit-margin analysis.

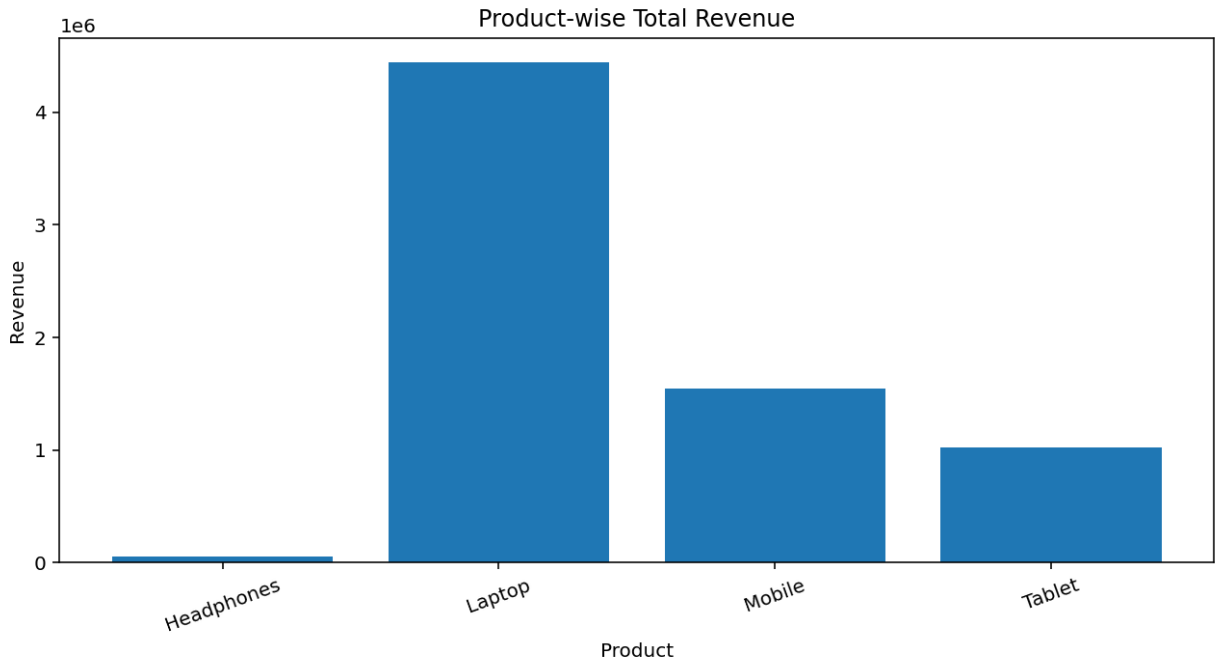
Python libraries such as Pandas, NumPy, and Matplotlib were used.

7. DATA VISUALIZATION

Five visualizations were created using Matplotlib.

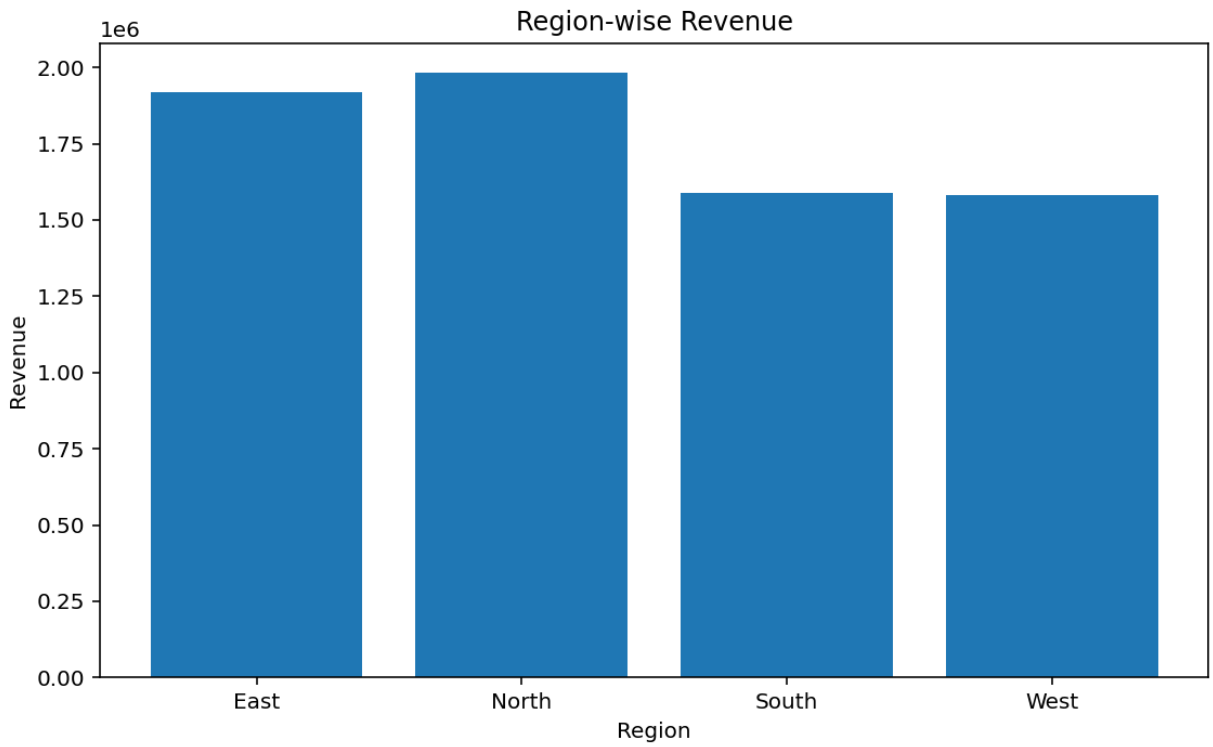
7.1 Product-wise Total Revenue

This chart compares revenue generated by different products and helps identify the highest-revenue product.



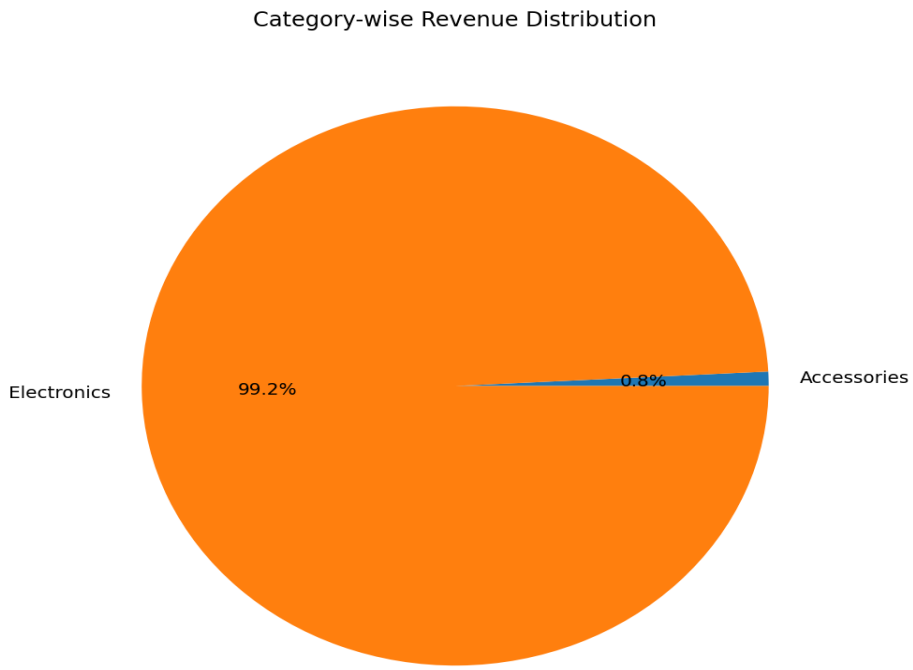
7.2 Region-wise Revenue

This visualization compares sales revenue across North, South, East, and West regions.



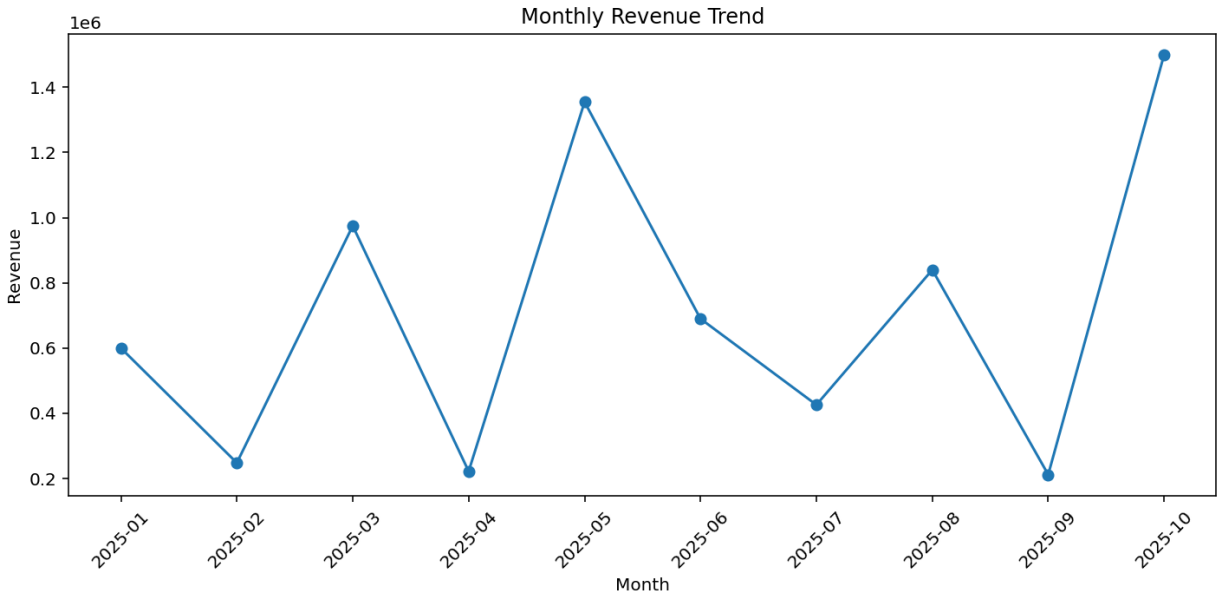
7.3 Category-wise Revenue Distribution

The pie chart represents the contribution of Electronics and Accessories to total revenue.



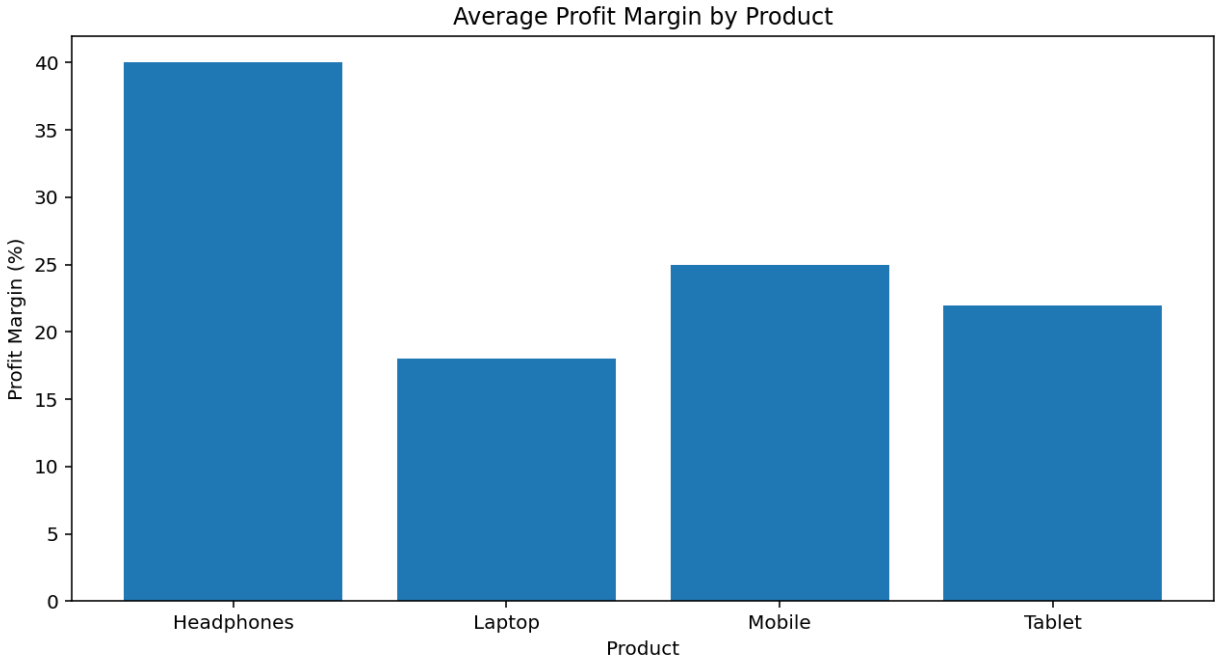
7.4 Monthly Revenue Trend

The line chart displays monthly revenue movement from January to October 2025.



7.5 Average Profit Margin by Product

This chart compares estimated average profit margins across the available products.



8. POWER BI DASHBOARD

An interactive dashboard was developed using Microsoft Power BI.

The dashboard contains three KPI cards:

1. Total Revenue: ₹70,68,000
2. Total Estimated Profit: Approximately ₹14 lakh
3. Average Estimated Profit Margin: 25.15%

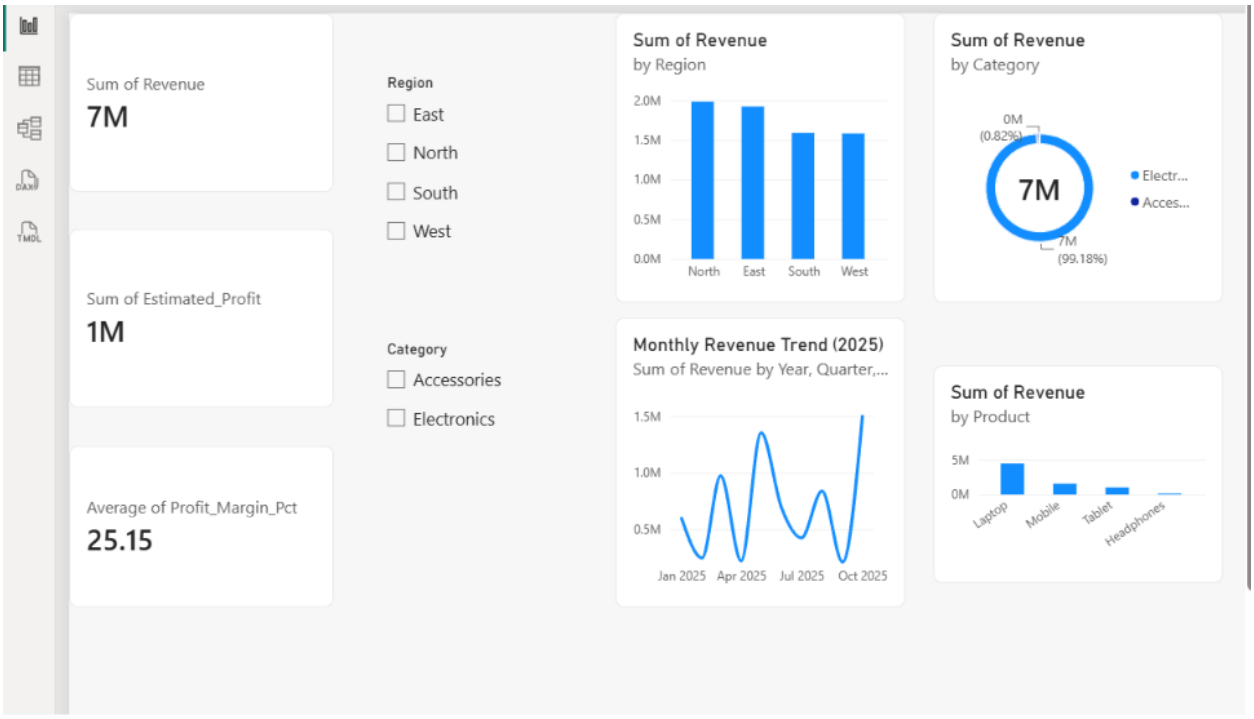
Dashboard visualizations:

- Product-wise Revenue
- Region-wise Revenue
- Monthly Revenue Trend
- Category-wise Revenue Distribution

Interactive slicers:

- Region
- Category

The slicers allow users to filter the dashboard and examine business performance for selected regions and product categories.


Sum of Revenue
7M
Sum of Estimated_Profit
1M
Average of Profit_Margin_Pct
25.15
Region
East
North
South
West
Category
Accessories
Electronics
Sum of Revenue
by Region
2.0M
1.5M
1.0M
0.5M
0.0M
North
East
South
West
Sum of Revenue
by Category
0M
(0.82%)
7M
7M
(99.18%)
Electr...
Acces...
Monthly Revenue Trend (2025)
Sum of Revenue by Year, Quarter,...
1.5M
1.0M
0.5M
Jan 2025
Apr 2025
Jul 2025
Oct 2025
Sum of Revenue
by Product
5M
0M
Laptop
Mobile
Tablet
Headphones

9. KEY BUSINESS INSIGHTS

Based on the analysis, the following insights were identified:

1. Product Performance

Laptops generate the highest revenue among the products in the dataset.

2. Regional Performance

The North region generates the highest revenue compared with the other regions.

3. Category Performance

Electronics contributes the largest share of total revenue.

4. Monthly Revenue Movement

Revenue fluctuates significantly across the analyzed months, indicating variations in sales performance.

5. Estimated Profitability

Headphones have the highest estimated profit margin under the assumed product cost structure.

10. BUSINESS RECOMMENDATIONS

The following recommendations can help improve business performance:

1. Inventory and Product Strategy

Focus on high-revenue products such as laptops while maintaining sufficient inventory and monitoring product demand.

2. Monthly Sales Improvement

Analyze lower-revenue months and introduce targeted marketing campaigns, promotional offers, and customer engagement activities to improve sales performance.

11. GIT AND GITHUB IMPLEMENTATION

Git and GitHub were used to manage the project and maintain version history.

The following activities were completed:

- Installed and configured Git.
- Created a GitHub account.
- Created a public repository named Sales-Revenue-Profit-Analysis.
- Initialized a local Git repository.
- Added project files using Git.
- Created meaningful commits.
- Connected the local repository to GitHub.
- Uploaded project files using Git Push.
- Created project documentation through README.md.

GitHub Repository:

https://github.com/krishdadwal07-ops/Sales-Revenue-Profit-Analysis

12. PROJECT STRUCTURE

The final project is organized into the following folders:

- Dataset – Extended sales dataset and documentation.
- Python Files – Data preparation and EDA scripts.
- Visualizations – Five generated charts.
- Power BI Files – Interactive dashboard.
- Report – Final project report.
- Screenshots – Project screenshots.
- Presentation – Final project presentation.

13. LIMITATIONS OF THE PROJECT

The dataset contains 20 illustrative sales records.

The original dataset did not contain actual transaction dates or product costs. Therefore:

- Dates were added for monthly analysis.
- Product cost rates were assumed for estimated cost calculations.
- Estimated profit and profit margins are illustrative.
- The results should not be interpreted as actual company financial performance.

A larger dataset containing verified transaction dates and actual product costs would improve the accuracy and reliability of future analysis.

14. CONCLUSION

The project successfully demonstrates the application of data analytics techniques to sales performance analysis.

Python was used for data preparation, exploratory data analysis, statistical calculations, and visualization. Microsoft Power BI was used to develop an interactive dashboard containing important performance indicators and business charts.

Git and GitHub were used for version control, project management, and documentation.

The project provided practical experience in data analytics, business intelligence, data visualization, and collaborative development workflows.

Overall, the project demonstrates how structured data analysis can help identify business patterns, evaluate sales performance, and support informed decision-making.

15. REFERENCES

1. Python Documentation – https://docs.python.org/3/
2. Pandas Documentation – https://pandas.pydata.org/docs/
3. NumPy Documentation – https://numpy.org/doc/
4. Matplotlib Documentation – https://matplotlib.org/stable/
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6. Git Documentation – https://git-scm.com/doc
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